

Supplementary Material S2

Synthetic Data Generator – Code & Parameterisation Details

Manuscript Title: HIMS CDI: A Governance Specification and Assurance Architecture for CTI-Driven Cybersecurity Risk Assessment in Healthcare

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Repository: <https://github.com/oluseun-akeredolu/HIMS-CDI>

Zenodo archive: <https://zenodo.org/records/16897633>

S2.1 Purpose

This supplement provides the complete code and parameterisation details for the synthetic data generator used to produce the 240 environment corpus for the specification-fidelity evaluation described in Section 4.1 of the main manuscript. All parameters are derived from publicly available healthcare incident statistics. The generator's output can be exactly reproduced using the fixed random seeds and correlation structures archived in the Zenodo repository.

The generator produces a synthetic corpus ($n = 240$), split into training (60 %, 144 environments), validation (20 %, 48 environments), and test (20 %, 48 environments) sets. Each environment is characterised by 20 indicators across five groups: CTI modality, semantic enrichment, fusion strategy, regulatory traceability, and explainability. Continuous features are perturbed with Gaussian noise ($\sigma = 0.10$) to simulate realistic measurement uncertainty, and labels are assigned deterministically according to the HIMS-CDI governance gates.

S2.2 Parameterisation Methodology

The synthetic corpus is constructed in four steps:

1. **Marginal distributions** – For each of the 20 indicators, marginal distributions (log-normal, beta, categorical) are specified based on the public statistics listed in Table S2.1.
2. **Dependency structure** – A latent multivariate normal framework with a target correlation matrix (derived from sector-level co-occurrence frequencies) captures inter-indicator dependencies.
3. **Latent-to-marginal transformation** – Latent normal samples are mapped to the target marginals using the probability integral transform.
4. **Label generation** – After computing CDI, CCR, latency, and memory from the indicator values, each environment receives a deterministic DEPLOY_SYNTH / NOT_DEPLOY_SYNTH label by applying the governance gates ($CDI \geq 0.70$, $CCR \geq 0.80$, latency < 200 ms, memory < 100 MB). The labelling logic is included in the full pipeline archived in the Zenodo repository.

The full correlation matrix is provided in the file `correlation_matrix.csv`. The exact parameter values (means, standard deviations, shape parameters) are documented in the `synthetic_data_generator.py` script archived in the Zenodo repository.

S2.3 Data Source Traceability Table

All parameters are extracted from the following publicly accessible sources (detailed extraction methodology in S2.1_Data_Source_Traceability.docx):

Source	Publisher	Year	Key Parameters Extracted
Verizon 2025 DBIR (Healthcare Sector)	Verizon Business	2025	Incident proportions, attack pattern frequencies, system intrusion rate
UK Cyber Security Breaches Survey 2024	Department for Science, Innovation and Technology	2024	Incident prevalence, detection latency estimates, control maturity
NCSC Annual Review 2024	UK National Cyber Security Centre	2024	Threat categories, severe incident counts, sector weights
NHS Digital Cyber Security in the NHS 2023–24	NHS Digital	2024	Incident classifications, reporting frequency, resilience maturity
Cyber Resilience of the Health and Care Sector 2024	Department of Health and Social Care	2024	Sector risk priorities, resilience indicators
ENISA Threat Landscape for the Health Sector	ENISA	2023	Healthcare threat taxonomy, attack vectors, likelihood rankings
IBM X-Force Threat Intelligence Index	IBM Security	2024	Breach vectors, dwell-time distributions, attack frequency
FBI IC3 Report	FBI	2024	Cybercrime categories, healthcare complaint counts
ICO Data Security Incident Trends	UK ICO	2024	Reported breach types, sector distribution
Health-ISAC Cybersecurity Survey	H-ISAC	2024	Control adoption rates, detection capability
MITRE ATT&CK v14.1 Technique Frequency Tables	MITRE Corporation	2024	Technique occurrence frequency, tactic distribution

No real NHS incident data are included; all statistics are aggregated public summaries.

S2.4 Generator Code (Reference Implementation)

The following Python script (synthetic_data_generator.py) is a reference implementation that reproduces the correlation structure of the 240-environment corpus. The exact version used in the evaluation, which applies the full empirical marginal transformations and stratified splitting, is archived at the Zenodo DOI.

```
python
#!/usr/bin/env python3
```

''''

Synthetic Data Generator – Reference Implementation (Main Manuscript v1.0)

This script illustrates the methodology used to generate the synthetic dataset for the HIMS-CDI specification-fidelity evaluation (Section 4.1).

The actual dataset is provided in the repository at `data/synthetic_hims_cdi_dataset.csv`.

This reference implementation reproduces the correlation structure but uses simplified marginal distributions for illustrative purposes. The original generator code, which relied on empirical distributions from public healthcare statistics, is versioned in the GitHub repository.

Dependencies: numpy, pandas, scipy

''''

```
import numpy as np
```

```
import pandas as pd
```

```
from scipy.stats import multivariate_normal
```

```
RANDOM_SEED = 42
```

```
np.random.seed(RANDOM_SEED)
```

```
N_ENVIRONMENTS = 240    # Total environments (see Section 4.1)
```

```
N_INDICATORS = 20
```

```
indicator_names = [
```

```
    "cti_modality_ind1", "cti_modality_ind2", "cti_modality_ind3", "cti_modality_ind4",
```

```
    "semantic_enrichment_ind1", "semantic_enrichment_ind2", "semantic_enrichment_ind3",  
    "semantic_enrichment_ind4",
```

```
    "fusion_strategy_ind1", "fusion_strategy_ind2", "fusion_strategy_ind3", "fusion_strategy_ind4",
```

```
    "regulatory_traceability_ind1", "regulatory_traceability_ind2", "regulatory_traceability_ind3",  
    "regulatory_traceability_ind4",
```

```

    "explainability_ind1", "explainability_ind2", "explainability_ind3", "explainability_ind4"
]

# Load target correlation matrix (provided in the repository)
corr_matrix = np.loadtxt("correlation_matrix.csv", delimiter=",")
assert corr_matrix.shape == (N_INDICATORS, N_INDICATORS), "Correlation matrix shape mismatch"

# Generate latent multivariate normal samples
latent = multivariate_normal.rvs(mean=np.zeros(N_INDICATORS), cov=corr_matrix,
size=N_ENVIRONMENTS)

# Transform to uniform marginals using the standard normal CDF
uniform = 0.5 * (1 + np.erf(latent / np.sqrt(2)))

# Apply simplified marginal transformations (illustrative)
# In the actual generator, these are mapped to the empirical distributions
# documented in the traceability table and the repository scripts.
data = uniform # placeholder; real generator applies complex transforms

df = pd.DataFrame(data, columns=indicator_names)

# Stratified 60/20/20 split (example – the actual implementation uses
# sklearn.model_selection.StratifiedShuffleSplit to preserve the proportion
# of DEPLOY_SYNTH labels in each split)
train = df.iloc[:144]
val = df.iloc[144:192]
test = df.iloc[192:]

train.to_csv("synthetic_train.csv", index=False)
val.to_csv("synthetic_val.csv", index=False)
test.to_csv("synthetic_test.csv", index=False)

```

```
print("Reference splits (train/val/test) generated.")
```

```
print("For the exact dataset used, refer to data/synthetic_hims_cdi_dataset.csv.")
```

Note on stratification: In the actual pipeline the train/validation/test split is stratified on the DEPLOY_SYNTH label to preserve the class proportions observed in the full corpus (67 % negative, 33 % positive). The reference implementation uses a simple sequential split for brevity; the full stratified splitting logic is available in the archived Zenodo scripts.

Execution:

```
python synthetic_data_generator.py
```

After running the full pipeline (including marginal transformations, label generation, and stratified splitting), the resulting test set will contain exactly 48 environments with 32 NOT_DEPLOY_SYNTH and 16 DEPLOY_SYNTH, matching the description in Section 4.1 of the main manuscript.

S2.5 Dataset Description

The final synthetic dataset comprises **240 environments**, each described by 20 numeric indicators and a binary risk class (DEPLOY_SYNTH or NOT_DEPLOY_SYNTH). The 60/20/20 stratified split yields **48 test environments** (32 negative, 16 positive), exactly as reported in the main manuscript (Section 4.1).

File: synthetic_hims_cdi_dataset.csv (full 240-row version) is available in the project repository at the Zenodo DOI.

S2.6 Reproducibility

All random seeds, correlation matrices, and environment configurations are archived in the permanent Zenodo repository (DOI: 10.5281/zenodo.16897633). A single-command script that regenerates the entire synthetic corpus and runs the full evaluation pipeline is included in the same archive.

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